Cauchy Distribution

1 Definition

A continuous random variable X is said to have a Cauchy distribution if its PDF is given by,

$$f_X(x) = \frac{\sigma}{\pi\left(\sigma^2 + (x-\mu)^2\right)}, \quad x \in \mathbb{R}.$$

Notation: $X \sim C(\mu, \sigma)$

Now, we will verify whether it is a PDF or not.

$$\begin{aligned}
\int_{-\infty}^{\infty} f(x)dx &= \int_{-\infty}^{\infty} \frac{1}{\pi}\frac{\sigma}{\sigma^2 + (x-u)^2}dx = \frac{1}{\pi}\sigma\int_{-\infty}^{\infty}\frac{1}{\sigma^2+z^2}dz \qquad \begin{matrix} x-u=z \\ dz=dx. \end{matrix} \\
&= \frac{\sigma}{\pi}\int_{-\infty}^{\infty}\frac{1/\sigma^2}{1+\left(\frac{z}{\sigma}\right)^2}dz \\
&= \frac{\sigma}{\pi}\frac{1}{\sigma^2}\int_{-\infty}^{\infty}\frac{1}{1+\left(\frac{z}{\sigma}\right)^2}dz. \quad \frac{z}{\sigma} = u \quad \Rightarrow dz = \sigma du \\
&= \frac{1}{\pi}\frac{1}{\sigma}\sigma\int_{-\infty}^{\infty}\frac{1}{1+u^2}du \\
&= \frac{1}{\pi}\left[\tan^{-1}u\right]_{-\infty}^{\infty} \\
&= \frac{1}{\pi}\left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right] = \frac{1}{\pi}\pi = 1.
\end{aligned}$$

2 CDF

$$\begin{aligned}
F_X(x) = \int_{-\infty}^{x} f_X(t)dt &= \int_{-\infty}^{x}\frac{\sigma}{\pi\left(\sigma^2+(t-\mu)^2\right)}dt \\
&= \frac{1}{\pi}\left[\tan^{-1}\left(\frac{t-\mu}{\sigma}\right)\right]_{-\infty}^{x} \\
&= \frac{1}{\pi}\left[\tan^{-1}\left(\frac{x-\mu}{\sigma}\right) - (-\pi/2)\right] \\
&= \frac{\pi}{\pi}\tan^{-1}\left(\frac{x-\mu}{\sigma}\right) + \frac{1}{2} = \frac{1}{2} + \frac{\tan^{-1}}{\pi}\left(\frac{x-\mu}{\sigma}\right), \quad x \in \mathbb{R}.
\end{aligned}$$

3 Median

$$\begin{aligned}
&F_X(\text{me}) = \frac{1}{2} \\
\Rightarrow &\frac{1}{\pi}\tan^{-1}\left(\frac{\text{me}-\mu}{\sigma}\right) = 0 \\
\Rightarrow &\tan^{-1}\left(\frac{\text{me}-\mu}{\sigma}\right) = 0. \\
\Rightarrow &\text{me} - \mu = 0 \Rightarrow \text{me} = \mu.
\end{aligned}$$

Remark

$$f_X(\mu - h) = \frac{\sigma}{\pi\left(\sigma^2 + h^2\right)} = f_X(\mu + h) \cdot \quad \forall h.$$

$\Rightarrow f_X(x)$ is symmetric about μ.
$\Rightarrow f_X(x)$ is symmetric about its median μ.

4 Mode

$f_X(x)$ is maximum iff $\sigma^2 + (x-u)^2$ is minimum iff $(x-\mu)^2 = 0$ iff $x = \mu$.
So, μ is the mode of the distribution.

5 Standard Cauchy Distribution

If $X \sim C(\mu, \sigma^2)$, the standardized random variable

$$Y = \frac{X - \mu}{\sigma} \sim C(0, 1).$$

CDF of Y,

$$\begin{aligned} F_Y(y) = P(Y \leqslant y) &= P\left[\frac{X-\mu}{\sigma} \leqslant y\right] \\ &= P[X \leqslant \mu + \sigma y] \\ &= f_X(\mu + \sigma y) \end{aligned}$$

PDF of Y is $\frac{d}{dy} f_Y(y) = f_X(\mu + \sigma y)\sigma = \sigma \frac{\sigma}{\pi(\sigma^2 + (\sigma y)^2)} = \frac{1}{\pi(1+y^2)}, y \in \mathbb{R}$,

6 Non-Existence of Moments

Let $X \sim C(0,1)$,

$E|X| = \int_{-\infty}^{\infty} |x| \frac{1}{\pi\left(1+x^2\right)} dx = \frac{2}{\pi} \lim_{a\to\infty} \int_0^a \frac{x}{(1+x^2)} dx.$

$$\begin{aligned} &= \frac{2}{\pi}\frac{1}{2} \lim_{a\to\infty} \left[\log\left(1+x^2\right)\right]_0^a \\ &= \frac{1}{\pi} \lim_{a\to\infty} \left[\log\left(1+a^2\right)\right] = \infty. \end{aligned}$$

Hence, $E|X|$ does not exist and consequently the higher order moments do not exist.

1. For a Cauchy distribution, show that $\mu_r' = E(X^r)$ exists iff $-1 < r < 1$.

$$\begin{aligned} E|X|^n &= \frac{2}{\pi} \int_0^{\infty} x^n \frac{1}{(1+x^2)} dx; \quad \begin{matrix} x^2 & = z. \\ 2xdx & = dz. \end{matrix} \\ &= \frac{2}{\pi}\frac{1}{2} \int_0^{\infty} \frac{z^{(n-1)/2}}{(1+z)} dz \\ &= \frac{1}{\pi} \int_0^{\infty} \frac{z^{\frac{n+1}{2}-1}}{(1+z)^{\frac{n+1}{2}+\frac{(1-n)}{2}}} dz \\ &= \frac{1}{\pi} B\left(\frac{n+1}{2}, \frac{1-n}{2}\right), \end{aligned}$$

which is a beta integral of second kind and is convergent iff

$$\begin{aligned} &\frac{n+1}{2} > 0, \frac{1-n}{2} > 0 \\ &\text{iff } -1 < r < 1 \end{aligned}$$

So, for $C(0,1)$ distribution, moments exists for $-1 < r < 1$.

7 Derivation of Cauchy PDF

A straight line AB is allowed to move through a fixed point A with co-ordinate $(0,0)$ and the length of the intersection (x) it makes with the X axis is noted. AB makes an angle $\theta = \angle OAP$ with the Y axis. Assuming θ has an Uniform distribution between $-\pi/2$ to $\pi/2$. Find the distribution of X.

$\angle OAP = \theta$ and $\theta \sim U(-\pi/2, \pi/2)$.

The PDF of $\theta = f(\theta), \begin{cases} \frac{1}{\pi}, & -\pi/2 < \theta < \pi/2 \\ 0 & \text{o.w.} \end{cases}$

From $\triangle OAP$, $\frac{OP}{OA} = \tan\theta \Rightarrow \frac{x}{\sigma} = \tan\theta \Rightarrow x = \sigma\tan\theta$.

The CDF of X is,

$$\begin{aligned} &-\pi/2 < \theta < \pi/2 \\ &-\infty < \tan\theta < \infty \\ &-\infty < \sigma\tan\theta < \infty \\ &-\infty < x < \infty \end{aligned}$$

$$\begin{aligned} G(x) &= P[X \leqslant x] \\ &= P[\sigma\tan\theta \leqslant x] \\ &= P\left[\theta \leqslant \tan^{-1}\left(\frac{x}{\sigma}\right)\right] \\ &= F_\theta\left(\tan^{-1}\left(\frac{x}{\sigma}\right)\right) \end{aligned}$$

The PDF of X is,

$$\begin{aligned} \frac{d}{dx}G(x) &= \frac{d}{dx}\left(F_\theta\left(\tan^{-1}\left(\frac{x}{\sigma}\right)\right)\right) \\ &= f_\theta\left(\tan^{-1}\left(\frac{x}{\sigma}\right)\right)\frac{1}{1+\left(\frac{x}{\sigma}\right)^2}\left(\frac{1}{\sigma}\right) \\ &= \frac{1}{\pi}\frac{\sigma^2}{\sigma^2+x^2}\frac{1}{\sigma} \\ &= \frac{\sigma}{\pi\left(\sigma^2+x^2\right)} \end{aligned}$$

$X \sim C(0,\sigma)$ (simply more the center to μ) $\rightarrow$ to get $c(\mu,\sigma)$.

2. If $X \sim C(0,1)$, show that $Y = \frac{1}{X} \sim C(0,1)$.

$$\begin{aligned} P[Y \leqslant y] = P\left[\frac{1}{X} \leqslant y\right] = P\left[\frac{1}{Y} \leqslant x\right] = P\left[X \geqslant \frac{1}{y}\right] &= 1 - P\left[X \leqslant \frac{1}{y}\right] \\ &= 1 - F_X\left(\frac{1}{y}\right) \end{aligned}$$

PDF of Y,

$$\begin{aligned} \frac{d}{dy}f_Y(y) = -f_X\left(\frac{1}{y}\right)\left(-\frac{1}{y^2}\right) &= \frac{1}{\pi\left(1+\frac{1}{y^2}\right)} \times \frac{1}{y^2} \\ = \frac{1}{\pi\left(1+y^2\right)}, y \in \mathbb{R}. \end{aligned}$$

$\therefore Y \sim C(0,1)$.

8 Comparison of Cauchy and Normal Distribution

Let $X \sim N(0,1)$ and $Y \sim C(0,1)$,

$$f_X(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}, \quad t \in \mathbb{R}$$
$$f_Y(t) = \frac{1}{\pi(1+t^2)}, \quad t \in \mathbb{R}$$

i) Both X and Y have symmetric distribution about 0. For Normal distribution, mean = median = mode = 0. But for Cauchy, median = mode = 0 but mean does not exist.

ii) $\text{Max}\, f_X(t) = f_X(0) = \dfrac{1}{\sqrt{2\pi}}, \quad \text{Max}\, f_Y(t) = \dfrac{1}{\pi}$

$$\pi > 2 \Rightarrow \pi^2 > 2\pi$$
$$\Rightarrow \frac{1}{\sqrt{2\pi}} > \frac{1}{\pi}$$

iii) For $|t| \to \infty$, $f_X(t)$ decreases to 0 rapidly than $f_Y(t)$.